

Gemini CLI

Modification

CISC 322

Group 3



 **Video URL:** [A3 Video](#)

 **Website:** [Group 3 Website](#)

Group Members



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Team Lead + SAAM Analysis



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Presenter
Impacts on High
Level and Low
Level
Architecture



Enrong
Pan

Presenter
SAAM Analysis



Omar

Use Cases



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Oleka

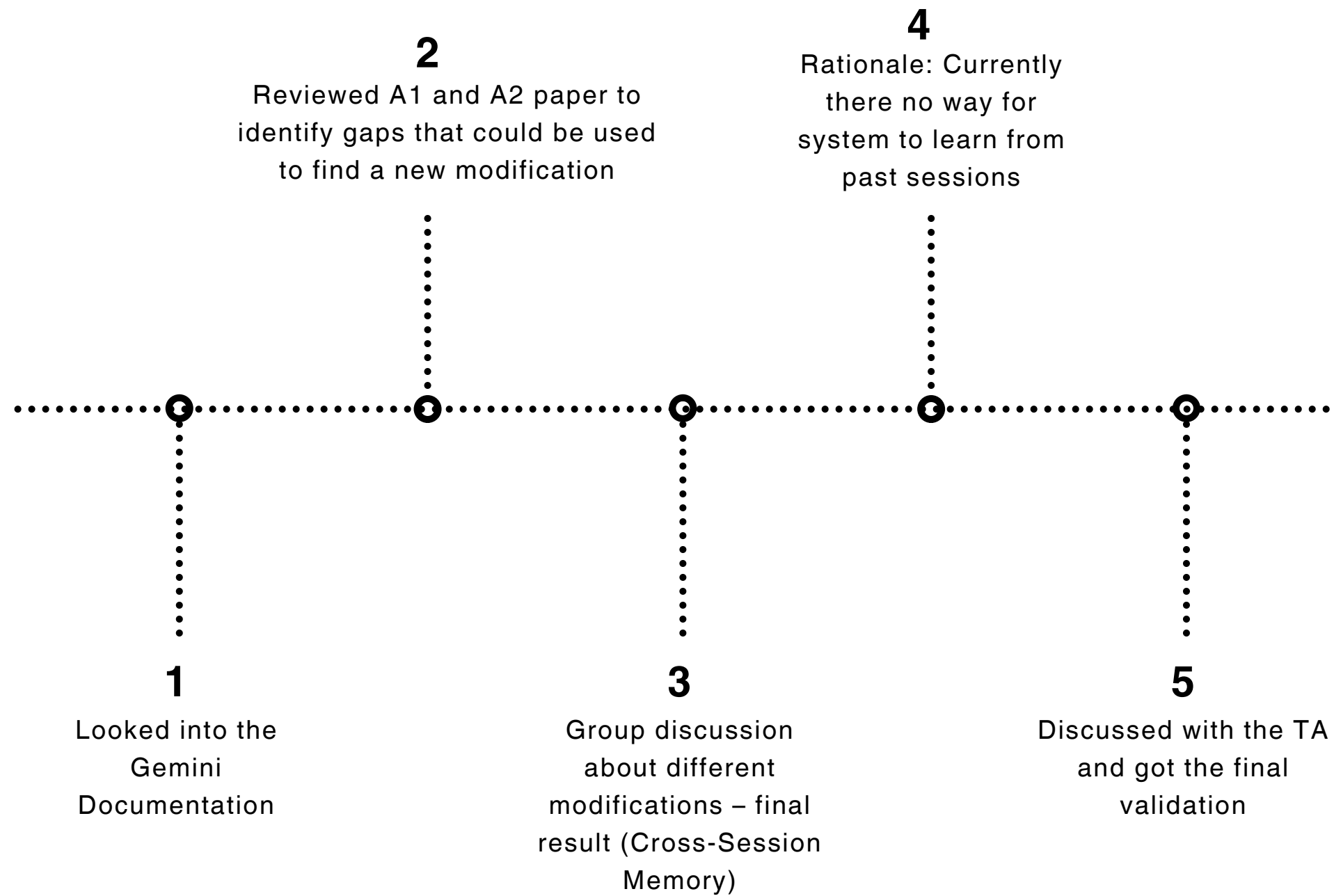
Abstract + Intro
+ Conclusion



Nandan
Bhut

Potential Risks + AI

Derivation and Motivation

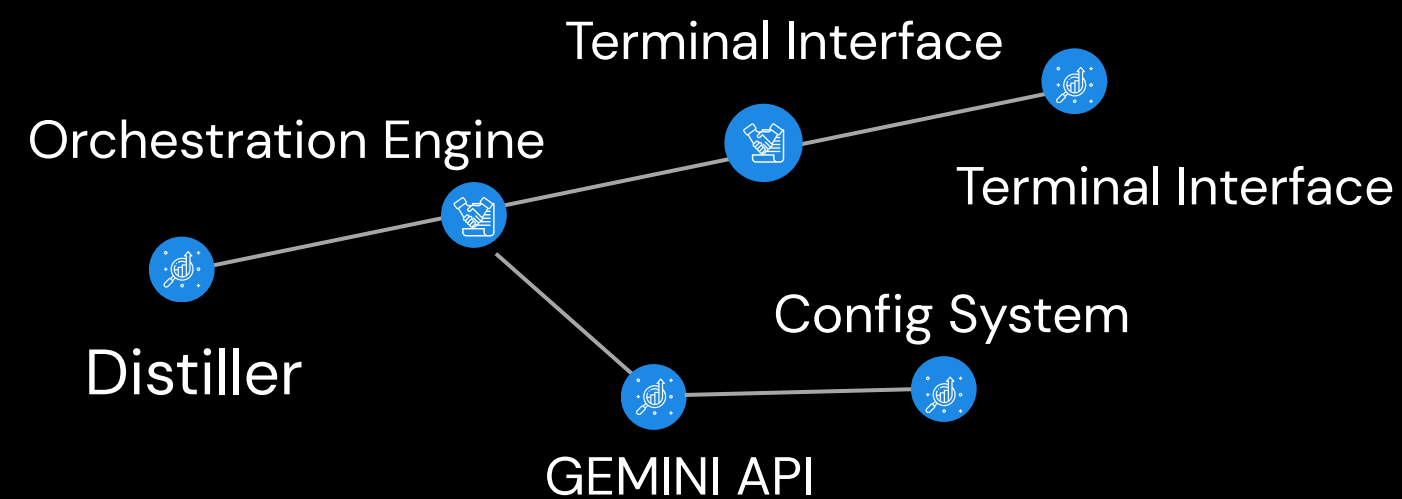


Implementation Approaches

1

Option A: Trace Distiller

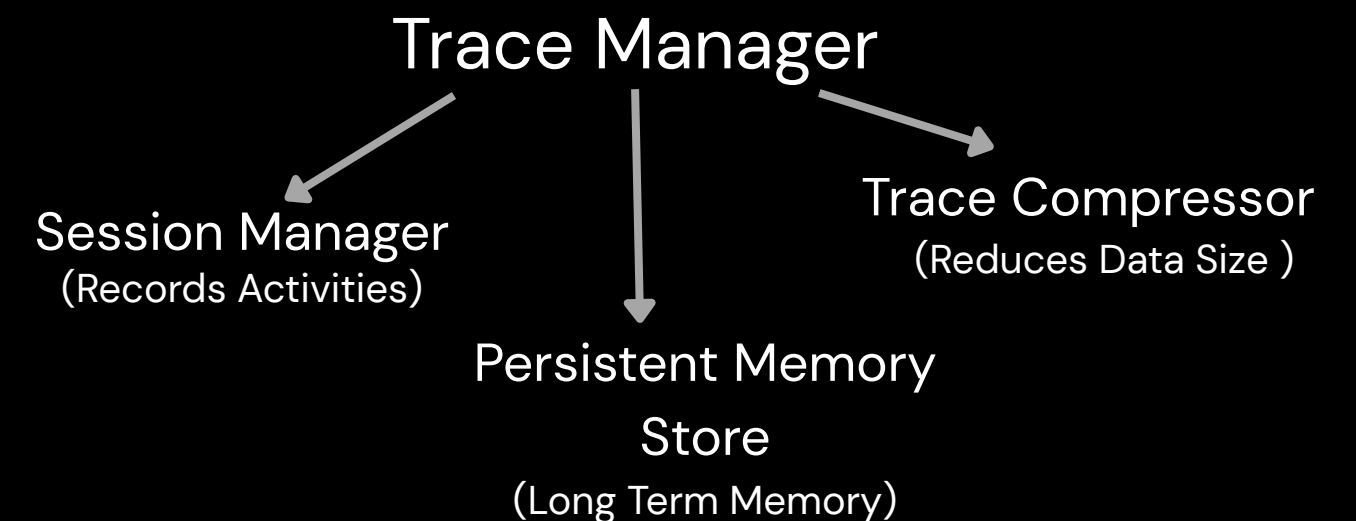
- Every Subsystem adds a Distiller component
- Rule-based heuristic, Independent component
- OE bundles summarized sessions with user feedback sent to API for final distillation
- Newly constructed memory sent to the config layer



2

Option B: Trace Manager

- Trace Manager spans multiple layers like config.
- Trace Manager made of 3 components
- Session Manager record activity other subsystems
- Trace Compressor process and compress data
- Persistent Memory Store stores long-term memory



SAAM Analysis

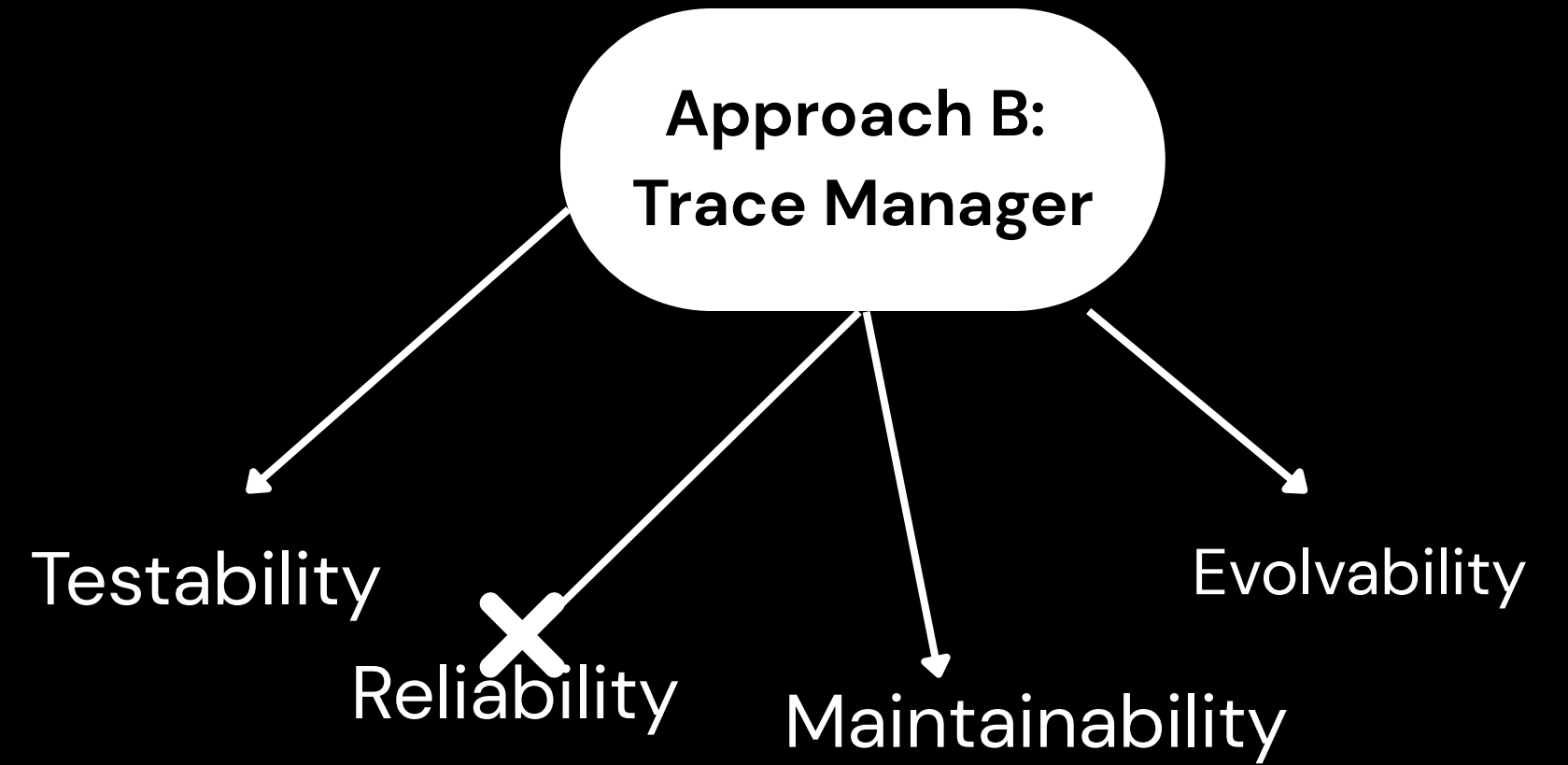
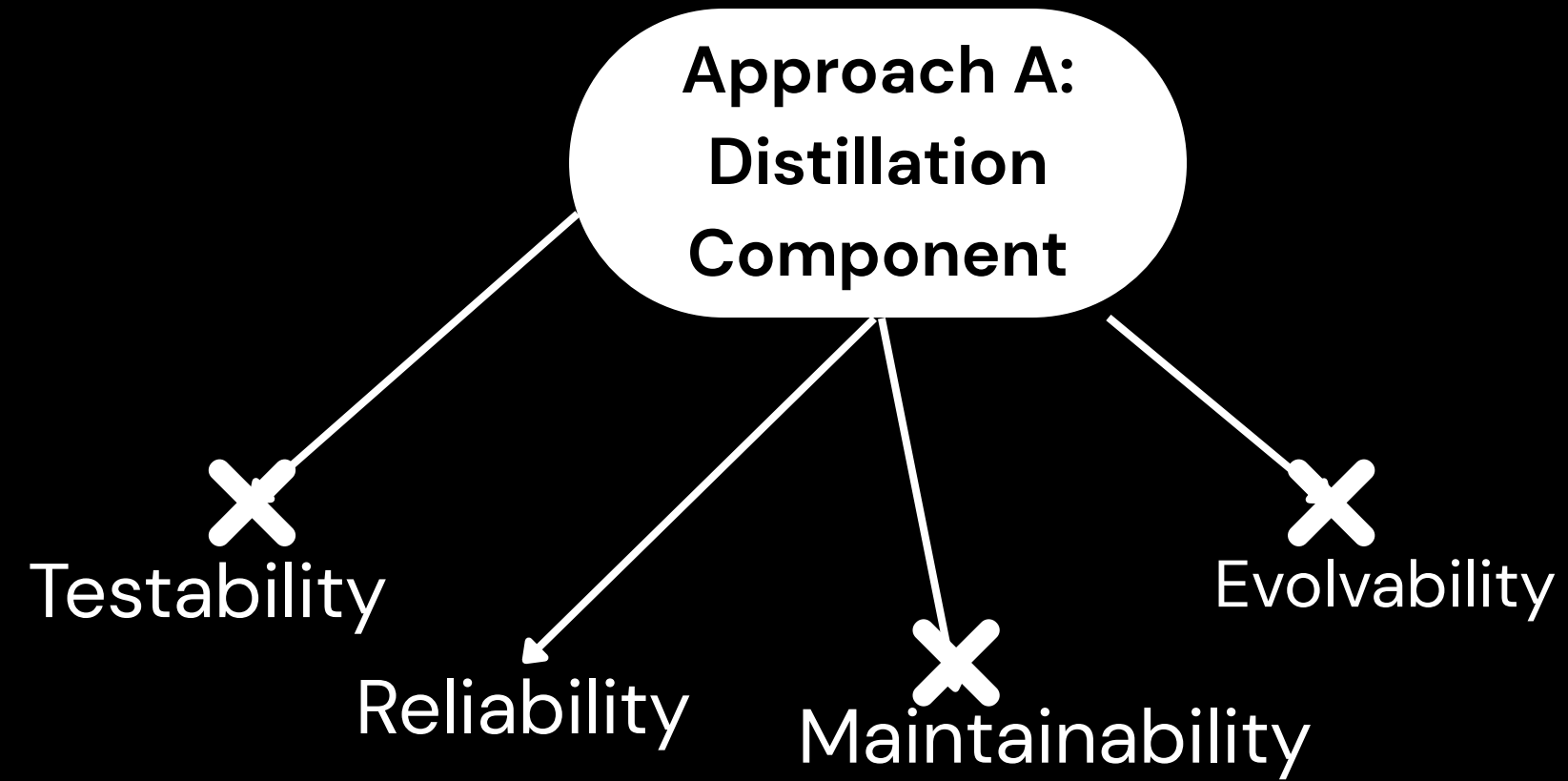
Stakeholders



Who: Devs maintaining CLI
NFR Priority: Maintainability & Testability
Value: Evolution, Operational Cost,



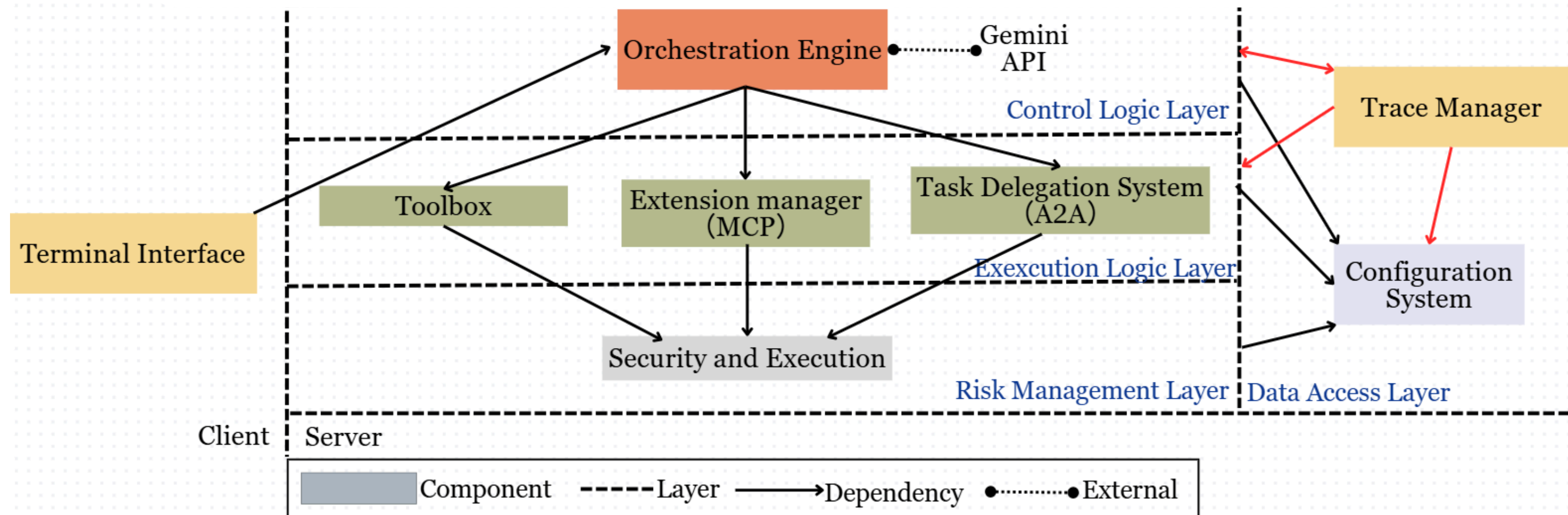
Who: Users who rely on CLI for AI-assisted coding
NFR Priority: Performance & Responsivness
Value: Privacy, supervision, and Authority over kept information



NFR Comparisons



Enhanced Impacts

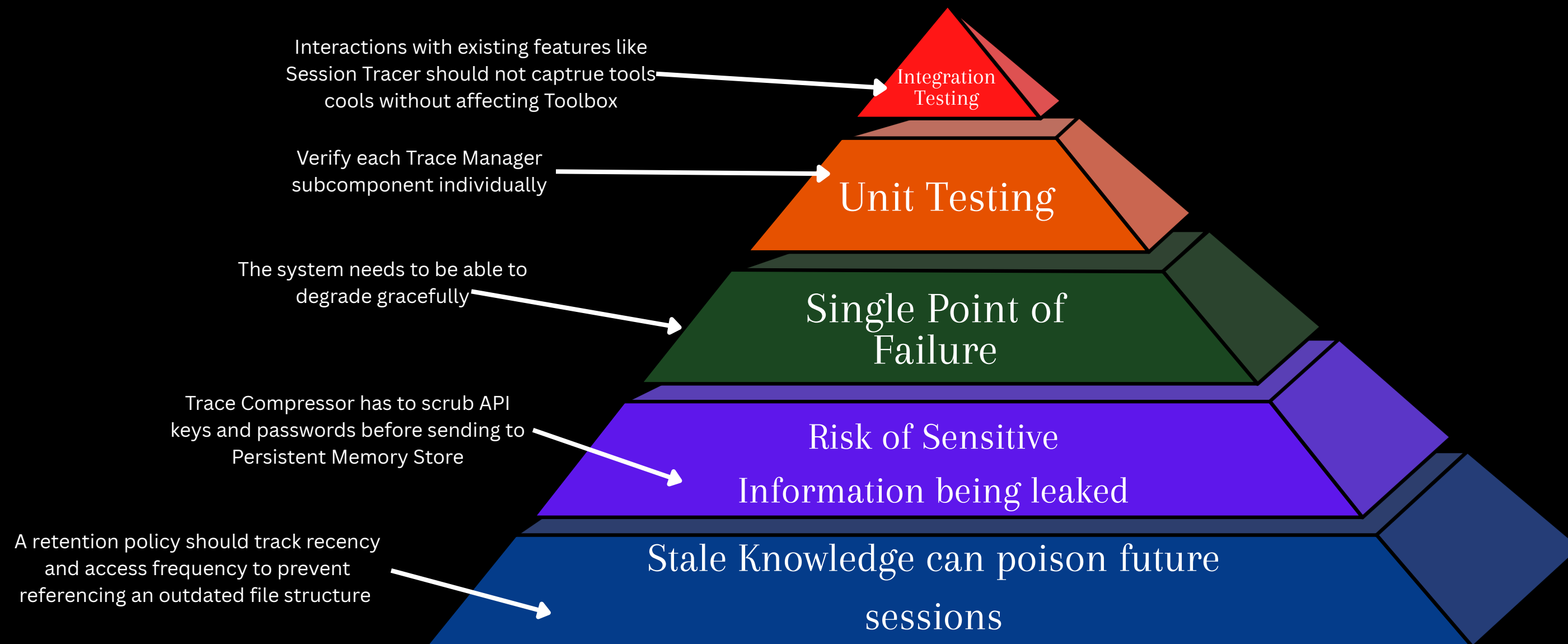


agent-session.ts

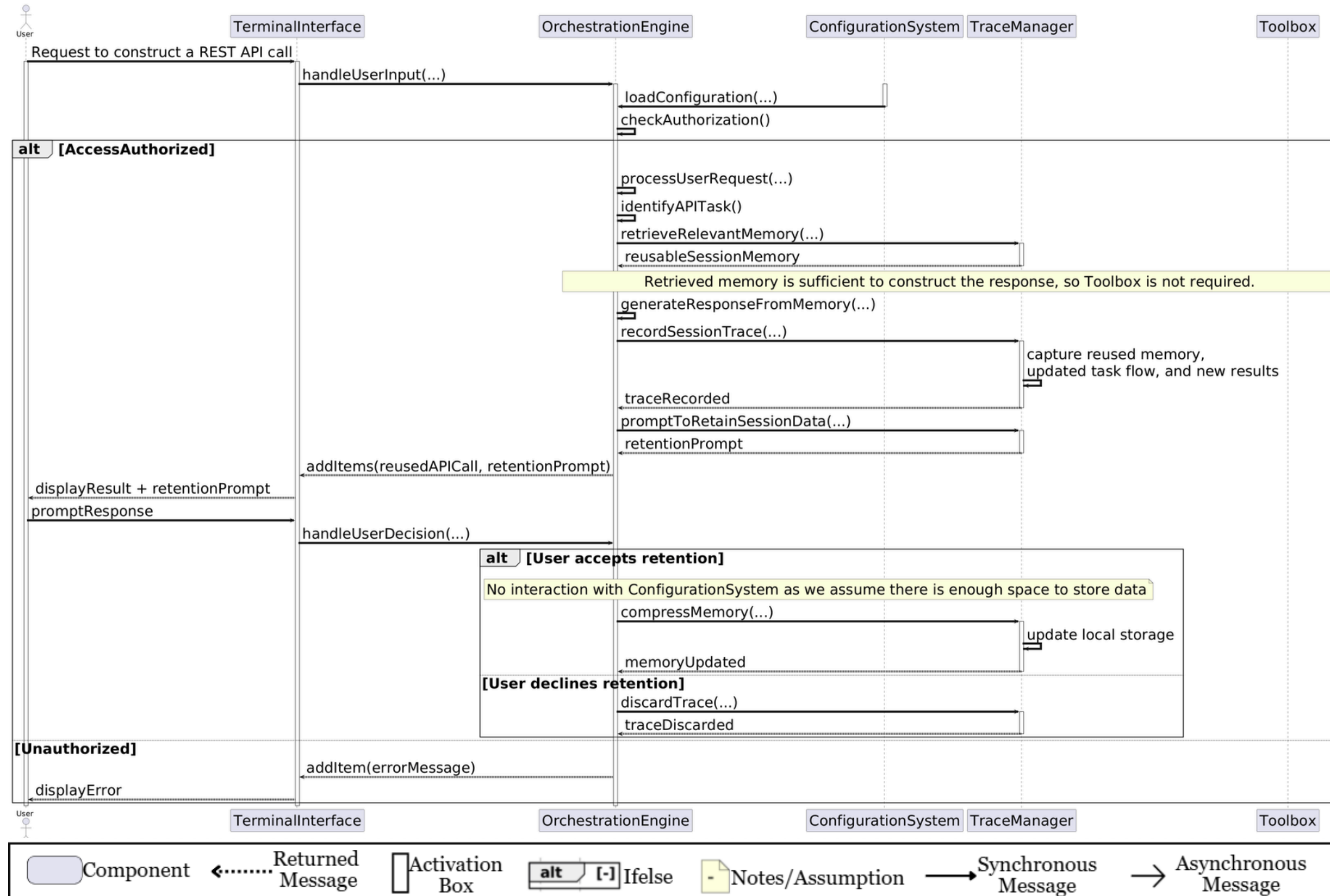
agent-scheduler.ts

types.ts

RISKS and Testing????



Use Case



AI Collaboration

Role of AI in A3: 5-7%

1. Brainstorming
2. Enhancement Ideas
3. Identifying Risks
4. Catching Details

Hey Gemini you are to act as an AI assistant that is an expert in System Designing and software architecture. Your job is to answer the User prompt to the best of your capability (you can use knowledge sources) and remain as faithful as possible. The user question is that they are wondering what sort of Advancements would bringing a Trace Manager have on the current concrete architecture. You are to break down the question into subparts: Number one is How would it affect the following seven subsystems: Orchestration Engine, Toolbox, Terminal Interface, Configuration System, Security and Execution, Task 2 Delegation, Extension Manager. You may use past information provided in regards to Assignment 1 and Assignment 2 answer the following task. When thinking of a solution you are to use ToT or CoT to think of various different solutions and use Marginalize and Popular Vote to come up with final task. You should also provide explanation and justifications to help the user keep track. The second subtask is that when adding the Trace Manager how would the NFRs be affected in a SAAM analysis. You just need provide a brief overview for SAAM Analysis to get the User started with thinking of ideas. The final output should be in table of N x 2 column where each table should represent a particular topic the user is interested in from the prompt

Trace Manager into the Gemini CLI concrete architecture, using a Chain of Thought (CoT) approach. By evaluating multiple options, the **Centralized Logging, Middleware Interceptors**, and "Popular Vote" for a robust system like Gemini favors the chosen solution. This minimizes latency in the ReAct loop while maximizing

This advancement impacts the existing subsystems and the requirements (NFRs).

ms

and functional changes required across the seven core components of the Trace Manager.

Architectural Impact & Advancement

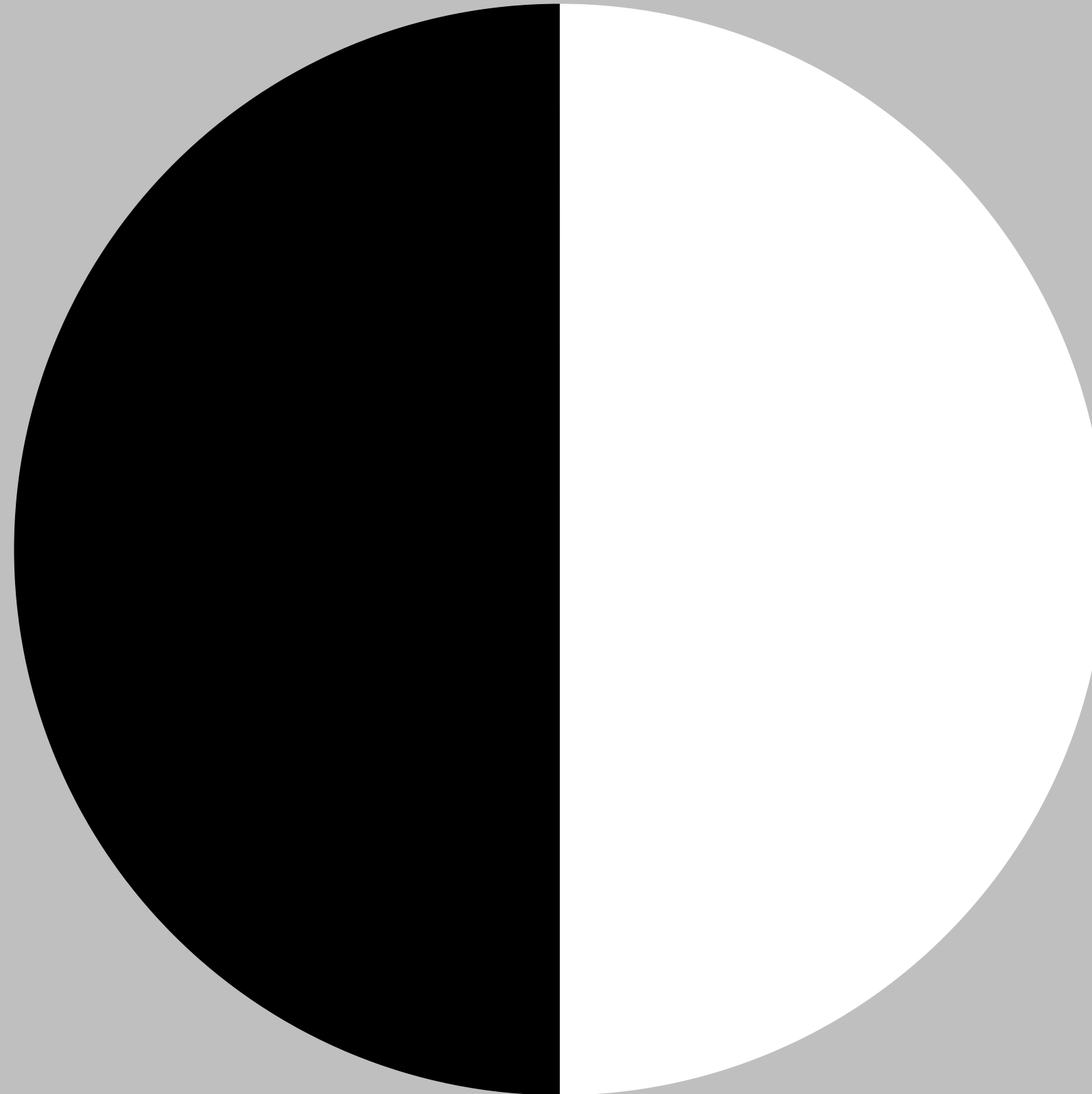
Acts as the primary "Subject." The Engine must now emit lifecycle events (e.g., `thought_start`, `action_selected`) to the Trace Manager to map the ReAct control flow.

Gains "Wrapper" logic. Every tool execution is wrapped to capture inputs, outputs, and execution time, providing a granular look at how external functions contribute to the final answer.

Receives a new "Debug/Trace" stream. The UI can now toggle between a clean user output and a "live trace" view, improving the developer experience during local testing.

New schema entries are required to define trace levels (e.g., `TRACE`, `DEBUG`, `TRACE`) and output destinations.

- Lesson 1: No perfect implementation for any software f...
- Lesson 2: Respect Existing Architecture instead restart...



Lesson 1: No perfect
implementation for any
software feature tradeoffs
exists
50%

Lesson 2: Respect
Existing Architecture
instead restarting
50%

Lessons Learned

Limitations:

**Theoretical analysis, not
physical!**

**Ability of Trace Manager
depends on GEMINI API**

In Conclusion...